DHARYATRA CHAUHAN

+91 9672077982 | chauhandharyatra@gmail.com | dharyatra.in | linkedin.com/in/dharyatra-chauhan | github.com/CDharyatra

PROFESSIONAL SUMMARY

AWS Certified Machine Learning Engineer - Associate with 2.6 years of experience building applied NLP, GenAI, RAG, and Python backend systems across enterprise analytics, text-to-SQL, document intelligence, and conversational AI. Experienced in designing production-oriented solutions with FastAPI, LangGraph, LangChain, pgvector, PostgreSQL, Docker, and AWS Bedrock, including authenticated APIs, RBAC, semantic retrieval, and session-aware workflows. Strong at translating business requirements into scalable, reliable AI applications and collaborating with engineering, product, and delivery teams.

TECHNICAL SKILLS

Programming: Python, SQL, C++

Generative AI & NLP: Retrieval-Augmented Generation (RAG), LangGraph, LangChain, Prompt Engineering, Text-to-SQL, Semantic Search, Vector Embeddings, Structured LLM Outputs, Conversational AI, AWS Bedrock, Ollama

Machine Learning & Data: Scikit-learn, XGBoost, Pandas, NumPy, PyTorch, TensorFlow**,** Feature Extraction, Data Preprocessing, Data Mining, Model Evaluation, Keyword Matching, Confidence Scoring

Backend & APIs: FastAPI, Flask, REST APIs, WebSockets, API Integration, Webhook Handling, JWT Authentication, Role-Based Access Control (RBAC), Microservices

Databases & Vector Search: MySQL, PostgreSQL, Pgvector, FAISS, Redis/Valkey, DynamoDB, Amazon OpenSearch

Cloud & Infrastructure: Amazon S3, AWS Lambda, Amazon API Gateway, Amazon SQS, Docker, Docker Compose, Linux

Engineering & Observability: Git, Pytest, CI/CD, Plotly, Streamlit, Typer, Prometheus, CloudWatch, CloudTrail

CERTIFICATIONS

AWS Certified Machine Learning Engineer

PROFESSIONAL EXPERIENCE

CLOUDTHAT TECHNOLOGIES — Bangalore, Karnataka

Machine Learning Engineer **Jan 2026 – Present**

Enterprise NLP Analytics Platform | FastAPI, LangGraph, PostgreSQL, Pgvector, Redis/Valkey, Plotly

- Built a production NLP analytics backend for a leading oil-and-gas enterprise using FastAPI and LangGraph, supporting TB-scale data and approximately 2,000 analytics queries per day across multiple business domains.
- Developed authenticated REST APIs for conversational analytics, session management, feedback collection, and vector-database operations, enabling reliable integration with enterprise workflows.
- Implemented a pgvector-backed RAG pipeline using schema metadata, few-shot examples, and KPI formulas to ground text-to-SQL generation and improve query relevance.
- Engineered JWT authentication, persona-based RBAC, and area-level data filtering so SO/DO/RO/FO users accessed only authorized records; built conversation memory, persistence, and Plotly visualization workflows with resilient retrieval fallbacks.

GenAI-Powered LMS Chatbot | AWS Bedrock, Lambda, S3, SQS, API Gateway, OpenSearch, DynamoDB

- Contributed to the architecture and implementation of a cloud-native, LMS-integrated GenAI chatbot for context-aware academic Q&A over medical course PDFs, lecture notes, and videos.
- Designed a serverless content-ingestion workflow using Amazon S3, API Gateway, AWS Lambda, and SQS for secure content intake, metadata extraction, chunking, asynchronous processing, and embedding generation.
- Developed the RAG design using Amazon Titan Text Embeddings v2 and Amazon OpenSearch to support semantic retrieval and course-module-aware responses, with multi-turn context persisted in DynamoDB.
- Defined security and observability requirements covering IAM, API authentication, KMS/TLS encryption, CloudWatch monitoring, CloudTrail auditing, and client handover documentation.

NEXGEN AI SOLUTION — Remote

Machine Learning Engineer **Dec 2025 – Jan 2026**

- Engineered Typer-based CLI scaffolding for managing Model Context Protocol (MCP) servers within the Conda ecosystem, structuring reusable command workflows and lifecycle operations that reduced server setup time by 30%.

- Designed and implemented a plugin architecture that integrated natural-language assistance and MCP server management into the core Conda CLI, defining extensible command boundaries and interfaces across two core capabilities.
- Implemented intent-to-command translation for more than 15 Conda operations and built robust subprocess error handling, validation, and reporting paths that improved debuggability and reduced issue-resolution time by approximately 20%.

VSURE CONSULTANCY SERVICES — Mumbai, Maharashtra

Machine Learning Engineer **Mar 2024 – Nov 2024**

- Developed a machine-learning data-extraction pipeline for bank statements, applying preprocessing, feature extraction, and structured-field parsing to reduce manual data-entry time by 50% and improve extraction accuracy by 30%.
- Integrated the extraction solution into the existing financial data-analysis pipeline in collaboration with cross-functional teams, improving processing speed by 30% and overall accuracy by 25% through workflow and preprocessing refinements.

BENCITI TECHNOLOGIES — Bangalore, Karnataka

Software Development Intern **May 2022 – Jul 2022**

- Developed and tested core features for NUDGE, a C++ mathematics platform, improving performance and reliability through debugging and technical documentation.

PROJECTS

AI Resume Evaluator & Job Description Matcher | Python, Streamlit, LangChain, FAISS, Ollama, Docker | github.com/CDharyatra/Resume-Evaluation-AI-Project-Analysis

- Built a modular AI application that parses PDF, DOCX, and TXT resumes, compares them with job descriptions, and generates alignment analysis, missing-skill identification, and targeted improvement suggestions.
- Implemented structured LLM evaluation with LangChain and Pydantic validation, combining deterministic keyword matching with model-generated analysis and fallback logic for service unavailability.
- Developed a RAG layer using sentence-transformer embeddings and FAISS similarity search to retrieve company-specific job requirements and ground recommendations in relevant context.
- Delivered an interactive Streamlit interface with upload validation, evaluation history, Plotly visualizations, downloadable Markdown reports, Docker Compose services, Redis persistence, Prometheus monitoring, health checks, and automated tests.

AI Market Intelligence Dashboard | Python, Pandas, Scikit-learn, Gemini API, Streamlit, Plotly | github.com/CDharyatra/AI_Market_Intel

- Built a Python market-intelligence pipeline that normalizes and analyzes Google Play and iOS App Store datasets, producing cross-platform matching, category analytics, and competitive insights.
- Processed a documented dataset of 15,839 mobile apps and generated confidence-scored AI insights, D2C funnel metrics, SEO opportunities, ad concepts, content ideas, and landing-page recommendations.
- Developed a CLI and Streamlit dashboard with reusable CSV, JSONL, Parquet, and Excel loaders, automated visualizations, and PDF executive-report generation.
- Structured the application for repeatable execution through centralized configuration, environment-based secret management, dependency tooling, and documented end-to-end pipeline runs.

Voice AI Lead Qualification Agent | Python, FastAPI, OpenAI GPT-4, Vapi, Airtable, n8n | github.com/CDharyatra/Interior-Design-Voice-AI-Agent

- Built a Python/FastAPI voice-agent workflow that retrieves leads from Airtable, initiates outbound calls through Vapi, and records call outcomes and transcripts through validated webhooks.
- Validated Vapi webhook payloads with Pydantic and routed completed, no-answer, busy, and failed calls into structured processing and persistence workflows.
- Used GPT-4 to extract structured lead attributes including budget, location, timeline, project scope, property size, qualification, summary, and next steps from call transcripts.
- Implemented retry handling, attempt limits, scheduled polling, failure-state tracking, and Airtable updates; documented n8n workflows for scheduled lead retrieval, batch processing, call triggering, and response handling.

EDUCATION

BML Munjal University — Gurugram, Haryana

B.Tech in Computer Science **September 2024**